Settle the argument: action and reaction

14 I

DIRECTIONS · Shade the bubble of the person who is right. Use the swap-the-names rule.

ROUND 1

ANA
"The pair forces cancel, so the carts can't move."

BEN
"They act on different objects, so they can't cancel."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"In a truck vs. car crash, the truck pushes harder."

LU
"The forces are equal. The car accelerates more."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"Weight and the table's push are a 3rd-law pair."

RAJ
"Both act on the book. Weight's partner is the book pulling Earth."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: force pairs

Language goal: I can **explain** why pair forces never cancel.



A ⇄ B

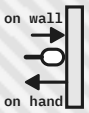
interaction

interacción · interação

two objects push or pull each other

my language: _____

use it: _____



→ ←

force pair

par de fuerzas · par de forças

equal size, opposite direction

my language: _____

use it: _____



force ON the box

F on B

on

sobre · sobre

the object that feels the force

my language: _____

use it: _____



F by A

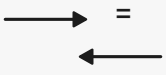
by

por · por

the object that exerts the force

my language: _____

use it: _____



=

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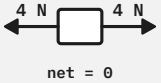
equal

igual · igual

same size

my language: _____

use it: _____



net = 0

+5 − 5 = 0

cancel

cancelar · cancelar

add to zero on ONE object

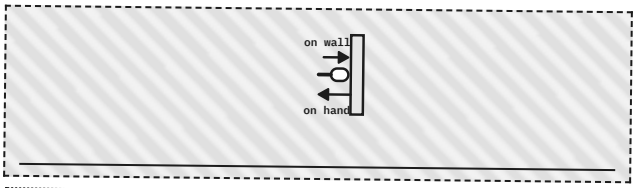
my language: _____

use it: _____

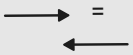
MATCH · WRITE THE WORD FOR EACH PICTURE



_____

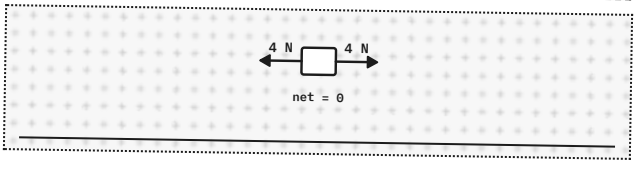


_____



=

_____



_____

I can point and say all 6 words to a partner □

R8 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Skateboard push-off, in six panels

14 E

DIRECTIONS · Two skaters push off each other. Draw both, with force arrows on EACH skater. Fill the captions.

1 · READY	2 · THE PUSH	3 · ANA
Both at rest. Masses: Ana 40 kg, Ben 80 kg.	Ana pushes Ben with 120 N. Ben pushes Ana with ____ N.	Ana's a = 120 ÷ 40 = ____ m/s².
4 · BEN	**5 · ROLLING APART**	**6 · THE TRUTH**
Ben's a = 120 ÷ 80 = ____ m/s².	Who moves faster? ____ Why?	Equal forces, ____ accelerations.

DRAW · ONE DIAGRAM PER SKATER (only forces ON that skater)

WHY DON'T THE PUSH FORCES CANCEL?

R8 1 2 3

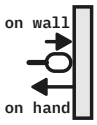
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: force pairs

Language goal: I can **swap the names** to find a partner force.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · A HAND PUSHES A WALL

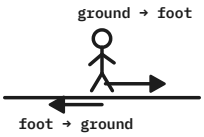


A ASKS "What is the partner force?"

B SAYS "I swap the names: the ____ pushes the ____, in the ____ direction."

A CHECKS "So the forces are ____ in size and act on ____ objects."

ROUND 2 · SWITCH · WALKING

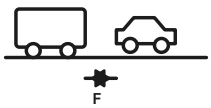


B ASKS "Your foot pushes the ground backward. What pushes you forward?"

A SAYS "The ____ pushes my ____ forward."

B ADDS "I want to add that this is why ____."

ROUND 3 · BOTH · TRUCK MEETS CAR



A "The forces are ____ because ____."

B "But the car's acceleration is ____ because its mass is ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Third-law forces never cancel because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R8 (1) (2) (3)

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Three third-law myths are on the loose

14 Y

Fill each poster: the truth and evidence that proves it.

WANTED

FOR CRIMES AGAINST PHYSICS

"The bigger object pushes harder."

LAST SEEN: a truck hitting a small car

THE TRUTH:

EVIDENCE:

REWARD: EQUAL FORCES

WANTED

FOR CRIMES AGAINST PHYSICS

"Action and reaction cancel out."

LAST SEEN: two carts pushing apart

THE TRUTH:

EVIDENCE:

REWARD: DIFFERENT OBJECTS

WANTED

FOR CRIMES AGAINST PHYSICS

"Weight and normal force are a pair."

LAST SEEN: a book on a table

THE TRUTH:

EVIDENCE:

REWARD: SWAP THE NAMES

CAPTURED! · WRITE YOUR OWN WANTED NOTICE

A wrong idea about today's physics I once believed:

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Pair or not a pair?

14 N

DIRECTIONS · Sort each pair of forces. Third-law pairs act on two different objects.

A hand pushes wall / wall pushes hand	**B** Earth pulls book / table pushes book	**C** foot pushes ground / ground pushes foot	**D** rope pulls box / friction on box
E Earth pulls Moon / Moon pulls Earth	**F** bat pushes ball / ball pushes bat	**G** rocket pushes gas / gas pushes rocket	**H** weight of a lamp / tension in its rope
I cart A pushes B / cart B pushes A	**J** you pull a sled / friction on the sled	**K** magnet pulls clip / clip pulls magnet	**L** swimmer pushes water / water pushes swimmer

3rd-LAW PAIR	NOT A PAIR	NOT SURE YET
A on B ⇄ B on A	same object, or different types	explain below

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

R8 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else